

Support Vector Machine (SVM)



What is SVM?

Support Vector Machine (SVM) is a **supervised machine learning algorithm** used for:

- Classification
- Regression
- Pattern recognition
- Image classification
- Face detection

SVM tries to find the **best boundary (hyperplane)** that separates different classes.

Key Idea of SVM

The main goal of SVM is:

Find the optimal hyperplane with the maximum margin between classes.

Important Terms

Term	Meaning
Hyperplane	Decision boundary separating classes
Margin	Distance between hyperplane and nearest points
Support Vectors	Closest data points to the hyperplane

Hyperplane Equation

$$w \cdot x + b = 0$$

Where:

Symbol	Meaning
(w)	Weight vector
(x)	Input feature vector
(b)	Bias term

Types of SVM

1. Linear SVM

Used when data is linearly separable.

Example:

- Two clearly separated classes

2. Non-Linear SVM

Used when data cannot be separated using a straight line.

Uses:

- Kernel Trick

Kernel Functions

1. Linear Kernel

$$K(x_i, x_j) = x_i^T x_j$$

2. Polynomial Kernel

$$K(x_i, x_j) = (x_i^T x_j + 1)^d$$

3. RBF Kernel

$$K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$$

4. Sigmoid Kernel

$$K(x_i, x_j) = \tanh(\alpha x_i^T x_j + c)$$

Working of SVM

Steps

1. Plot the training data
 2. Find possible hyperplanes
 3. Calculate margins
 4. Select hyperplane with maximum margin
 5. Classify new data points
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Example — Binary Classification



Suppose we have two classes:

- Red circles
- Blue squares

SVM finds the best line that separates them with maximum margin.

Advantages of SVM

- Works well with high-dimensional data
 - Effective for text classification
 - Memory efficient
 - Good accuracy
 - Works for both linear and non-linear data
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Disadvantages of SVM

- Slow for very large datasets
 - Hard to choose correct kernel
 - Complex mathematical calculations
 - Less effective with noisy data
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Applications of SVM

- Face Detection
 - Image Classification
 - Spam Detection
 - Handwriting Recognition
 - Bioinformatics
 - Text Categorization
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Comparison with Other Algorithms

Algorithm	Speed	Accuracy	Best For
SVM	Medium	High	Complex classification
Naive Bayes	Very Fast	Good	Text data
KNN	Slow	Medium	Small datasets
Decision Tree	Medium	Good	Rule-based problems

Python Example

```
from sklearn import svm

# Features
X = [[0, 0], [1, 1]]

# Labels
y = [0, 1]

# Create model
model = svm.SVC(kernel='linear')

# Train model
model.fit(X, y)

# Predict
prediction = model.predict([[2, 2]])

print(prediction)
```

Important Interview Questions

1. What is SVM?

A supervised learning algorithm used for classification and regression.

2. What is a Hyperplane?

A decision boundary separating classes.

3. What are Support Vectors?

Nearest data points to the hyperplane.

4. What is Kernel Trick?

Technique used to solve non-linear problems.

Short Notes

Definition

SVM is a supervised machine learning algorithm that finds the optimal hyperplane for classification.

Main Goal

Maximize the margin between classes.

Advantages

- High accuracy
- Effective in high dimensions
- Works with non-linear data

Disadvantages

- Slow for large datasets
- Complex calculations

Applications

- Image recognition
- Spam filtering
- Face detection
- Text classification